

Nervous System:

Structure & Function

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PHS 220 — Physiology

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§1 ORGANISATION OF THE NERVOUS SYSTEM

The nervous system is the body's primary communication, integration, and control network. Comprising over 100 billion neurons and trillions of supportive glial cells, it coordinates perception, movement, homeostasis, cognition, and behaviour. It is divided into two main structural components and several functional subdivisions.

Division	Components	Key Structures	Function
Central Nervous System (CNS)	Brain + Spinal Cord	Cerebrum, cerebellum, brainstem, diencephalon; cervical-sacral cord	Integration and processing; decision making; higher functions
Peripheral Nervous System (PNS)	12 pairs cranial nerves + 31 pairs spinal nerves + ganglia + peripheral nerve plexuses	Brachial plexus, lumbosacral plexus, cranial nerves I-XII	Communication between CNS and body (sensory input; motor output)
Somatic NS	Voluntary motor and conscious sensory	Corticospinal tract; dorsal column; α -motor neurons	Voluntary movement; conscious sensation
Autonomic NS (ANS)	Sympathetic + Parasympathetic + Enteric	Preganglionic and postganglionic neurons; ganglia; plexuses	Involuntary control of viscera, glands, smooth/cardiac muscle
Enteric NS	Intrinsic neural network of GI tract (Auerbach's and Meissner's plexuses)	Myenteric and submucosal plexuses	"Brain in the gut" — largely autonomous; controls GI motility, secretion

AFFERENT vs EFFERENT

Afferent (sensory): Carry signals **FROM** periphery **TOWARDS** the CNS. Cell body in dorsal root ganglion (DRG) for spinal; in cranial nerve ganglia for head. Sensory modalities: touch, pain, temperature, proprioception, special senses.

Efferent (motor): Carry signals **FROM** CNS **TOWARDS** periphery. Somatic efferents go directly to skeletal muscle (α -motor neurons). Autonomic efferents have a two-neuron chain (preganglionic → ganglion → postganglionic → effector).

§2 NEURONS — STRUCTURE & CLASSIFICATION

Neurons are the fundamental signalling units of the nervous system, each uniquely specialised for rapid electrical communication. Despite enormous morphological diversity, all neurons share a common set of structural components.

Part	Description	Function
Cell Body (Soma/Perikaryon)	Contains nucleus (large, pale, prominent nucleolus), Nissl substance (rough ER — highly active protein synthesis), Golgi apparatus, mitochondria	Biosynthesis of proteins, lipids, neurotransmitters; integration of synaptic inputs; trophic centre of neuron

Part	Description	Function
Dendrites	Multiple branching processes; covered in dendritic spines (postsynaptic contact points); tapered; no myelin	Receive synaptic inputs; summate EPSPs and IPSPs; transmit local graded potentials towards soma
Axon Hillock	Transition zone between soma and axon; lowest threshold for AP generation	Site of action potential initiation (highest density of voltage-gated Na ⁺ channels)
Axon (Nerve Fibre)	Single process; uniform diameter; may branch distally (collaterals); may be myelinated; can be >1 m long (e.g., sciatic nerve fibres)	Conducts action potentials from hillock to terminals; axoplasmic transport (anterograde → proteins/organelles to terminals; retrograde → recycled material back to soma)
Axon Terminal (Synaptic Knob / Bouton)	Contains synaptic vesicles loaded with neurotransmitters; abundant mitochondria; docked vesicles at active zones	Release neurotransmitter into synaptic cleft on arrival of action potential

2.1 Functional Classification of Neurons

Class	Structure	Location	Examples
Multipolar	Multiple dendrites + one axon from soma	Most neurons in CNS; α-motor neurons	Spinal motor neurons; cortical pyramidal cells; Purkinje cells (cerebellum)
Bipolar	One dendrite + one axon (one from each pole)	Special sense organs	Retinal bipolar cells; cochlear/vestibular neurons; olfactory receptor neurons
Pseudounipolar	Single process that bifurcates into central and peripheral branches	Dorsal root ganglia; trigeminal ganglion	All primary sensory neurons for somatic sensation; proprioception
Unipolar	Single process only (no separate axon and dendrite)	Invertebrate NS (rare in humans)	Mostly invertebrates

2.2 Nerve Fibre Classification (Erlanger-Gasser)

Fibre Type	Myelination	Diameter	Conduction Velocity	Function
Aα (Ia, Ib)	Heavily myelinated	12–20 μm	70–120 m/s	Proprioception (muscle spindle Ia; Golgi tendon Ib); α-motor (to extrafusal fibres)
Aβ (II)	Myelinated	6–12 μm	30–70 m/s	Mechanoreception (light touch, vibration, pressure)
Aγ	Myelinated	3–6 μm	15–30 m/s	Motor to intrafusal fibres (muscle spindle fusimotor)
Aδ (III)	Lightly myelinated	1–5 μm	5–30 m/s	Sharp/acute pain (first pain); cold temperature; crude touch

Fibre Type	Myelination	Diameter	Conduction Velocity	Function
B	Lightly myelinated	<3 μm	3–15 m/s	Autonomic preganglionic fibres
C (IV)	Unmyelinated	0.2–1.5 μm	0.5–2 m/s	Chronic/burning pain (second pain); warmth; postganglionic sympathetic fibres; itching

LOCAL ANAESTHETIC ORDER OF BLOCKADE

Local anaesthetics block sodium channels. They block smaller fibres before larger ones: C (pain/sympathetic) → A δ (acute pain/cold) → A β (touch/pressure) → A α (motor/proprioception). This is why patients undergoing spinal or epidural anaesthesia first notice loss of pain, then temperature, then touch, and motor paralysis last — with recovery in the reverse order.

§3 GLIAL CELLS

Glial cells outnumber neurons approximately 10:1 and provide essential structural, metabolic, and immunological support. Far from being passive 'filler', glia actively modulate synaptic transmission and are critically involved in neuropathology.

Glial Type	Location	Key Functions
Astrocytes	CNS (most abundant glia)	Structural scaffolding; form blood-brain barrier (endfeet around capillaries); K ⁺ spatial buffering; glutamate uptake (prevents excitotoxicity); glycogen storage; growth factor secretion; scar formation after injury (reactive astrogliosis)
Oligodendrocytes	CNS	Produce myelin sheaths for CNS axons (each myelinates multiple axons — up to 50 internodes); maintain axonal integrity. Destroyed in multiple sclerosis.
Schwann Cells	PNS	Produce myelin for PNS axons (one Schwann cell per internode — one axon only); guide nerve regeneration after injury; myelinated by wrapping around axon; produce neurotrophins (NGF, BDNF)
Microglia	CNS	Brain-resident macrophages (monocyte lineage); phagocytose debris, pathogens, synapses (synaptic pruning); produce inflammatory cytokines; activated in neuroinflammation; "surveillant" state in health → "reactive" in disease
Ependymal Cells	Lining of ventricles and central canal of spinal cord	Produce CSF (choroid plexus specialised ependyma); ciliated surface moves CSF; form blood-CSF barrier; ependymoma (tumour of ependymal cells)
Radial Glia	Developing brain	Neural progenitor cells; scaffold for neuronal migration during cortical development; give rise to astrocytes and oligodendrocytes postnatally

3.1 The Blood-Brain Barrier (BBB)

The BBB is a specialised system of tight junctions between cerebral capillary endothelial cells, reinforced by astrocyte endfeet, pericytes, and a thick basement membrane. It restricts passage of large molecules, charged substances, and most pathogens into the CNS, maintaining a stable neurochemical environment.

Crosses BBB Freely	Requires Specific Transport	Cannot Cross (without disruption)
O ₂ , CO ₂ , water, alcohols, many lipid-soluble drugs (e.g., anaesthetics, opioids)	Glucose (GLUT1 — essential for brain metabolism); amino acids (LAT transporters)	Most antibiotics, most chemotherapy agents; large proteins; immune cells (unless activated)

§4 THE NERVE IMPULSE: ACTION POTENTIAL

4.1 Resting Membrane Potential

The resting membrane potential (RMP) of a neuron is approximately **-70 mV** (inside negative relative to outside). This potential arises from the unequal distribution of ions across the membrane and the selective permeability of the membrane at rest.

Ion	Intracellular (mM)	Extracellular (mM)	Equilibrium Potential	RMP Contribution
K ⁺	~140	~4	-94 mV (E _K)	Primary determinant of RMP (membrane most permeable to K ⁺ at rest via leak channels)
Na ⁺	~10	~140	+60 mV (E _{Na})	Small contribution at rest (low permeability); drives depolarisation during AP
Cl ⁻	~4	~110	-70 mV (E _{Cl})	Roughly at equilibrium; important for GABA inhibition (hyperpolarisation)
A ⁻ (Organic anions)	Large negative proteins	~0	N/A	Cannot cross membrane; contribute to inside negativity

The Na⁺/K⁺-ATPase pump (3Na⁺ out : 2K⁺ in per cycle) maintains these gradients, making a small direct electrogenic contribution (-4 mV) and continually restoring ions after action potentials.

4.2 Action Potential Phases

Phase	V _m (approx.)	Ion Movement	Channel State
Resting	-70 mV	None net; K ⁺ leak maintains RMP	K ⁺ leak open; Na ⁺ channels closed; K ⁺ gated closed

Phase	V _m (approx.)	Ion Movement	Channel State
Depolarisation (Rising phase)	-70 → +30 mV (threshold ≈ -55 mV)	Na ⁺ rushes IN (E _{Na} = +60 mV drives inward current)	Voltage-gated Na ⁺ channels OPEN rapidly (activation gate opens)
Overshoot (Peak)	+30 to +40 mV	Na ⁺ entry continues briefly	Na ⁺ channels at peak activation; K ⁺ channels begin opening
Repolarisation (Falling phase)	+30 → -70 mV	K ⁺ rushes OUT (E _K = -94 mV drives outward current)	Na ⁺ channels INACTIVATE (h-gate closes); voltage-gated K ⁺ channels OPEN wide
After-Hyperpolarisation (AHP)	-70 to -80 mV	K ⁺ continues to leave briefly	K ⁺ channels closing slowly; Na ⁺ channels still inactivated → absolute refractory period ending → relative refractory period
Return to Rest	-80 → -70 mV	K ⁺ channels close; Na ⁺ channels recover (h-gate re-opens)	Na/K-ATPase restores gradients over time (but not needed for each single AP)

4.3 Refractory Periods & Propagation

Property	Definition	Mechanism	Significance
Absolute Refractory Period	No AP possible regardless of stimulus strength	Na ⁺ channels inactivated (h-gate closed); cannot reopen until membrane repolarises to ~-70 mV	Ensures AP propagates in one direction (prevents backward spread); limits maximum firing frequency (~500-1000 Hz)
Relative Refractory Period	AP possible but requires supra-threshold stimulus	Na ⁺ channels recovered but K ⁺ channels still partly open → membrane hyperpolarised (AHP)	Allows frequency modulation — stronger stimuli overcome AHP → higher firing rate
Saltatory Conduction	AP jumps from node to node in myelinated fibres	Myelin insulates internode; ion current flows only at Nodes of Ranvier; depolarises next node directly	Much faster conduction (up to 120 m/s); energy efficient (less membrane to repolarise)

§5 SYNAPTIC TRANSMISSION

5.1 Chemical Synapse — Steps

- **AP arrives at axon terminal** → membrane depolarisation
- **Voltage-gated Ca²⁺ channels (Cav2.1, 2.2) open** → Ca²⁺ influx into terminal (Ca²⁺ acts as the trigger for exocytosis)
- **Ca²⁺ binds synaptotagmin** on synaptic vesicle → conformational change in SNARE proteins (synaptobrevin + SNAP-25 + syntaxin) → vesicle fusion with presynaptic membrane →

exocytosis

- **Neurotransmitter released into synaptic cleft** (20–50 nm gap)
- **NT binds postsynaptic receptors** → opens ion channels (ionotropic) or activates signalling cascades (metabotropic/GPCRs)
- **NT removed from cleft** by: (a) Reuptake transporters (e.g., SERT for serotonin, DAT for dopamine, NET for NE, EAAT for glutamate); (b) Enzymatic degradation (AChE → ACh; MAO/COMT → catecholamines); (c) Diffusion

5.2 Key Neurotransmitters

Neurotransmitter	Synthesis	Receptor Types	Action	Key Pathways
Glutamate	From glutamine (glutaminase); from α -KG (TCA)	AMPA, NMDA, Kainate (ionotropic); mGluR (metabotropic)	Excitatory (major); NMDA critical for synaptic plasticity (LTP)	Corticospinal; thalamocortical; hippocampal; cerebellar; most excitatory synapses in CNS
GABA	Glutamate + GAD (glutamate decarboxylase); requires B6	GABAA (Cl^- ionotropic — BZDZ/barbiturate sites); GABAB ($\text{K}^+/\text{Ca}^{2+}$ GPCR)	Inhibitory (major); GABAA: $\uparrow \text{Cl}^- \rightarrow$ hyperpolarisation; GABAB: slow inhibition	Cerebellar Purkinje cells; striatum; cortical interneurons; basal ganglia output
Acetylcholine (ACh)	Choline + Acetyl-CoA (ChAT enzyme)	Nicotinic (nAChR — ionotropic, Na^+/K^+); Muscarinic M1–M5 (GPCR)	Excitatory at NMJ; varied CNS effects (arousal, attention, memory)	NMJ (all somatic motor); Autonomic ganglia (preganglionic); Basal forebrain → cortex/hippocampus (Meynert — memory); brainstem nuclei (REM sleep)
Dopamine	Tyrosine → L-DOPA (TH) → Dopamine (DOPA decarboxylase)	D1–D5 (all GPCRs)	Reward; motivation; movement control; inhibits prolactin	Mesolimbic (reward; addiction); Mesocortical (cognition, working memory); Nigrostriatal (movement — depleted in PD); Tuberoinfundibular (inhibits prolactin)
Serotonin (5-HT)	Tryptophan + TPH → 5-HTP → 5-HT	5-HT ₁ –7 (GPCRs mostly); 5-HT ₃ (ionotropic — nausea)	Mood; sleep; appetite; GI motility; nausea	Raphe nuclei → whole brain; GI tract (90% of body's serotonin); spinal dorsal horn (pain modulation)
Noradrenaline (NE)	Dopamine + D β H → NE	α 1, α 2, β 1, β 2, β 3 (all GPCRs)	Alertness; attention; fight-or-flight; pain modulation	Locus coeruleus → whole CNS (arousal, attention); SNS postganglionic
Glycine	Serine (via serine hydroxymethyltransferase)	GlyR (Cl^- ionotropic)	Inhibitory (primarily spinal cord and brainstem)	Spinal Renshaw cells (inhibit motor neurons); brainstem interneurons; blocked by strychnine → convulsions

Neurotransmitter	Synthesis	Receptor Types	Action	Key Pathways
Substance P	Preprotachykinin gene	NK1 receptor (GPCR)	Pain transmission (from C-fibre nociceptors to dorsal horn)	Dorsal horn pain pathways; NK1 antagonists for emesis (aprepitant)

§6 THE BRAIN — GROSS ANATOMY

Region	Divisions	Key Structures	Functions
Cerebrum (Telencephalon)	Two hemispheres; 4 lobes per hemisphere; connected by corpus callosum, anterior and posterior commissures	Cortex; white matter; basal ganglia; hippocampus; amygdala; olfactory bulbs	Highest neural functions: cognition, language, voluntary movement, sensation, memory, emotion
Diencephalon	Thalamus; Hypothalamus; Epithalamus (pineal); Subthalamus	Thalamic nuclei (relay and integrative); Hypothalamus (homeostasis); Pineal gland (melatonin)	Sensory relay to cortex (thalamus); homeostasis and ANS control (hypothalamus); circadian (pineal)
Mesencephalon (Midbrain)	Tectum (superior and inferior colliculi); Tegmentum; Cerebral peduncles	Superior colliculi (visual reflex); Inferior colliculi (auditory relay); Substantia nigra; Red nucleus; PAG; CN III, IV nuclei	Visual and auditory reflexes; pupillary light reflex; dopamine production; pain modulation
Pons	Basilar pons (transverse fibres to cerebellum); Pontine tegmentum	Locus coeruleus; Raphe nuclei; CN V, VI, VII, VIII nuclei; Pontine respiratory group	Relay to cerebellum; cranial nerve functions; respiratory rhythm
Medulla Oblongata	Pyramids (CST decussation); Inferior olive; Dorsal surface (4th ventricle floor)	Cardiovascular centre; Respiratory centre; Vomiting centre; CN IX–XII nuclei; Decussation of pyramids and medial lemniscus	Vital autonomic centres (HR, BP, breathing); cranial nerve functions; sensory/motor decussation
Cerebellum	2 hemispheres + vermis; 3 lobes; folia (convolutions)	Purkinje cells (output); dentate/interposed/fastigial nuclei (output); inferior/middle/superior cerebellar peduncles	Coordination; fine motor control; balance; timing of movements; motor learning; ipsilateral body representation

§7 CEREBRAL CORTEX — FUNCTIONAL AREAS

Area	Brodman #	Lobe	Function	Deficit if Damaged
Primary Motor Cortex	BA 4	Frontal (precentral gyrus)	Voluntary movement; somatotopic (homunculus); upper motor neuron origin of CST	Contralateral hemiparesis/hemiplegia; UMN signs (↑ tone, ↑ reflexes, +Babinski)
Premotor Cortex	BA 6 (lateral)	Frontal	Planning motor sequences; mirror neurons	Apraxia; loss of skilled voluntary movement patterns
Supplementary Motor Area (SMA)	BA 6 (medial)	Frontal (medial)	Bimanual coordination; initiation of self-generated movement	Difficulty initiating movement; alien hand syndrome
Primary Somatosensory Cortex	BA 1, 2, 3	Parietal (postcentral gyrus)	Conscious somatosensation; somatotopic (sensory homunculus)	Contralateral hemianesthesia; loss of fine touch, proprioception, 2-point discrimination
Primary Visual Cortex	BA 17	Occipital (calcarine sulcus)	Conscious visual perception; striate cortex	Contralateral homonymous hemianopia (with macular sparing from bilateral supply)
Primary Auditory Cortex	BA 41, 42	Temporal (Heschl's gyri)	Conscious sound perception; tonotopic organisation	Cortical deafness (bilateral lesion required for significant deficit)
Broca's Area	BA 44, 45	Frontal (inferior; usually left)	Speech production; motor planning of articulation	Broca's (expressive) aphasia: non-fluent speech; good comprehension
Wernicke's Area	BA 22	Temporal (superior; usually left)	Speech comprehension; language interpretation	Wernicke's (receptive) aphasia: fluent but meaningless speech; poor comprehension
Prefrontal Cortex	BA 9, 10, 11, 46, 47	Frontal (anterior)	Executive function; planning; working memory; personality; decision-making	Personality change; poor impulse control; working memory deficits; poor planning

CORTICAL LATERALISATION

Language (90% right-handers; 70% left-handers): Left hemisphere dominant for speech (Broca's) and language comprehension (Wernicke's). Arcuate fasciculus connects them — damage → conduction aphasia.

Visuospatial processing: Right hemisphere dominant — right parietal lesions → spatial neglect (ignores left half of world).

Hemispheric motor/sensory representation is contralateral (decussation in medulla/cord). Cerebellar representation is ipsilateral (double decussation).

§8 THE LIMBIC SYSTEM

The limbic system is a phylogenetically ancient network of interconnected structures surrounding the brainstem that mediates emotion, memory formation, olfaction, motivation, and homeostatic behaviours. It forms the interface between cortical cognition and hypothalamic autonomic/endocrine control.

Structure	Key Connections	Primary Functions	Lesion Effects
Hippocampus (CA1-CA4 + Dentate gyrus)	Entorhinal cortex (main input); mammillary bodies (via fornix); septum	Declarative memory (episodic + semantic) formation; spatial navigation (place cells, grid cells)	Bilateral destruction: profound anterograde amnesia (cannot form new declarative memories); H.M. case study
Amygdala	Cortex (all lobes); hippocampus; hypothalamus; thalamus; brainstem	Fear processing and conditioning; emotional salience; aggression; reward; autonomic fear responses (↑ HR, cortisol)	Klüver-Bucy syndrome (bilateral): placidity; hypersexuality; hyperorality; visual agnosia
Cingulate Gyrus (Anterior)	Prefrontal cortex; motor cortex; thalamus; amygdala	Error detection; attention; emotional regulation; pain affect component; motivated behaviour	Akinetic mutism (bilateral anterior cingulate lesions)
Mammillary Bodies	Hippocampus (fornix) → mammillary bodies → anterior thalamus (mammillothalamic tract) → cingulate	Memory relay (Papez circuit); orientation	Destroyed in Wernicke-Korsakoff syndrome (thiamine deficiency) → severe anterograde and retrograde amnesia; confabulation
Hypothalamus	Limbic system; brainstem; cortex; pituitary	Homeostasis; feeding; thirst; temperature; circadian; neuroendocrine control (via pituitary); autonomic regulation	Disrupted temperature regulation; hormonal imbalances; obesity (lateral hypothalamic damage)

§9 THE BASAL GANGLIA

The basal ganglia are a collection of subcortical nuclei involved in the modulation and selection of voluntary movements. They do not initiate movement but filter and refine motor programs, facilitating desired movements while suppressing unwanted ones. They also have important roles in cognition, habit learning, and reward processing.

Nucleus	Component	Neurotransmitter	Function
Striatum	Caudate nucleus + Putamen (main input nuclei)	GABA (output); receives dopamine from SNc and glutamate from cortex	Receives motor, cognitive, and limbic cortical inputs; gateway into basal ganglia circuit

Nucleus	Component	Neurotransmitter	Function
Globus Pallidus Interna (GPI)	Main output nucleus of basal ganglia	GABA (inhibitory output → thalamus)	Tonically inhibits thalamus; balanced by direct (disinhibition) and indirect (increased inhibition) pathways
Globus Pallidus Externa (GPe)	Indirect pathway component	GABA (inhibits STN)	Intermediate relay; inhibits subthalamic nucleus
Subthalamic Nucleus (STN)	Indirect pathway; key surgical target (DBS for PD)	Glutamate (excitatory output)	Activates GPI → inhibits thalamus; STN lesion → hemiballismus
Substantia Nigra pars compacta (SNc)	Dopaminergic input to striatum (nigrostriatal pathway)	Dopamine	Modulates striatal D1 (direct, excitatory) and D2 (indirect, inhibitory) receptors; depleted in Parkinson's disease
Substantia Nigra pars reticulata (SNr)	Additional output nucleus (like GPI)	GABA	Inhibitory output to thalamus; controls eye movements (to superior colliculus)

DIRECT vs INDIRECT PATHWAY

Direct pathway (GOES): Cortex → Striatum (D1 receptors) → GPI ↓ → Thalamus ↑ → Cortex ↑ → MOVEMENT FACILITATED. Dopamine excites D1 → facilitates movement.

Indirect pathway (STOPS): Cortex → Striatum (D2 receptors) → GPe ↓ → STN ↑ → GPI ↑ → Thalamus ↓ → Cortex ↓ → MOVEMENT SUPPRESSED. Dopamine inhibits D2 → reduces suppression.

Parkinson's Disease: ↓ Dopamine (SNc degeneration) → D1 understimulated (↓ direct) + D2 over-inhibited (↑ indirect) → net ↑ GPI inhibition of thalamus → BRADYKINESIA, RIGIDITY. Treatment with L-DOPA restores balance.

Huntington's Disease: Striatal (indirect pathway) neuron loss → ↓ GPI inhibition → ↑ thalamic output → CHOREA (excessive unwanted movement).

§10 SPINAL CORD — STRUCTURE & TRACTS

10.1 Cross-Sectional Anatomy

The spinal cord is organised around a central H-shaped grey matter region surrounded by white matter. The grey matter contains neuronal cell bodies; the white matter contains myelinated axons organised into columns (funiculi).

Grey Matter Region	Laminae (Rexed)	Neurons Present
Dorsal Horn (posterior)	I-VI	Sensory relay neurons; substantia gelatinosa (II — pain/temperature modulation); Clarke's column (III/IV/VII — proprioceptive relay to cerebellum)
Intermediate Zone	VII	Interneurons; Clarke's nucleus (C8-L3); IML (lateral horn, T1-L2 — sympathetic preganglionic neurons); Onuf's nucleus (S2-S4 — urethral and anal sphincter MNs)

Grey Matter Region	Laminae (Rexed)	Neurons Present
Ventral Horn (anterior)	VIII-IX	α -motor neurons (extrafusal fibres — voluntary movement); γ -motor neurons (intrafusal fibres — spindle sensitivity)
Central Canal Region	X	Commissural neurons; pain input

10.2 Major Ascending (Sensory) Tracts

Tract	Location	Modality	Decussation	Key Clinical Test
Dorsal Column-Medial Lemniscal (DCML)	Posterior funiculus (ipsilateral)	Fine touch; vibration; proprioception; 2-point discrimination; stereognosis	Medulla (sensory decussation of medial lemniscus)	128 Hz tuning fork; Romberg test; joint position sense
Spinothalamic Tract (Anterolateral)	Anterior and lateral funiculus (contralateral — decussates in cord)	Pain (A-delta + C fibre); temperature; crude touch; pressure	Spinal cord (within 1-3 segments of entry in anterior white commissure)	Pin prick; cold/warm temperature
Dorsal Spinocerebellar	Posterior lateral funiculus (ipsilateral)	Proprioception from lower limb (unconscious)	Does NOT decussate	Cerebellar ataxia if damaged
Cuneocerebellar	Lateral funiculus (ipsilateral)	Proprioception from upper limb (unconscious)	Does NOT decussate	Upper limb cerebellar signs

10.3 Major Descending (Motor) Tracts

Tract	Origin	Location	Function	Decussation
Lateral Corticospinal Tract (CST)	Primary motor cortex (BA4), premotor, SMA	Lateral funiculus (contralateral)	Voluntary fine motor movements (especially distal limbs); UMN for spinal LMNs	Pyramidal decussation in medulla (~85% of fibres)
Anterior (Ventral) CST	Primary motor cortex	Anterior funiculus (ipsilateral until decussation)	Axial and proximal limb muscles	Decussates in cord near target level
Rubrospinal Tract	Red nucleus (midbrain tegmentum)	Lateral funiculus	Limb flexor control; fine movements (minor in humans)	Decussates in midbrain

Tract	Origin	Location	Function	Decussation
Reticulospinal Tract	Pontine + medullary reticular formation	Anterior and lateral funiculi	Modulates reflex activity; tone; posture; autonomic control; gating of pain	Variable (both bilateral)
Vestibulospinal Tract	Lateral vestibular nucleus (Deiter's)	Anterior funiculus (ipsilateral)	Antigravity muscle tone; postural adjustments; extensor activation	Does NOT decussate
Tectospinal Tract	Superior colliculus	Anterior funiculus	Head/eye reflexes (orienting to visual/auditory stimuli)	Decussates in midbrain

BROWN-SÉQUARD SYNDROME (Hemisection of Spinal Cord)

Classic teaching case illustrating tract anatomy. Following hemisection at e.g. T10 (right side):
IPSI LATERAL (Right) side below lesion: Spastic paresis (CST — ipsilateral until medulla); loss of proprioception and fine touch (DCML — ipsilateral until medulla); spared pain/temperature (ALS decussated in cord).

CONTRALATERAL (Left) side 1-2 levels below lesion: Loss of pain and temperature (ALS — already crossed to left side in cord).

AT the level of lesion (ipsilateral): LMN signs (segmental); band of ipsilateral sensory loss (all modalities from entering dorsal root).

§11 THE AUTONOMIC NERVOUS SYSTEM (ANS)

The ANS controls involuntary functions of viscera, glands, smooth muscle, and cardiac muscle. Unlike the somatic system (single neuron from CNS to effector), the ANS uses a two-neuron chain: a **preganglionic neuron** (myelinated, B-fibre) from the CNS synapses in a **ganglion** onto a **postganglionic neuron** (unmyelinated, C-fibre) that reaches the effector.

Feature	Sympathetic NS	Parasympathetic NS
Outflow (CNS origin)	Thoracolumbar: T1-L2 (intermediolateral column)	Craniosacral: CN III, VII, IX, X; S2-S4
Preganglionic neuron	Short (synapses in paravertebral/prevertebral ganglia)	Long (synapses in or near target organ)
Postganglionic neuron	Long (travels to effector)	Short (within organ wall)
Ganglion location	Paravertebral chain (sympathetic trunk) OR prevertebral ganglia (coeliac, SMA, IMA)	Terminal ganglia (in/near organ: ciliary, pterygopalatine, otic, submandibular, cardiac, pulmonary, enteric)
Pre-ganglionic NT	ACh → nicotinic nAChR	ACh → nicotinic nAChR
Post-ganglionic NT (primary)	Noradrenaline (NE) → α and β receptors	Acetylcholine (ACh) → Muscarinic M1-M5

Feature	Sympathetic NS	Parasympathetic NS
Exceptions	Sweat glands (cholinergic sympathetic); Adrenal medulla (ACh → chromaffin cells → Adrenaline release)	CN III → pupillary constrictor, ciliary (M3); CN VII → lacrimal, nasal, submandibular; CN IX → parotid (M3); CN X → heart, lungs, GI (proximal)
Physiological role	"Fight, Flight, Fright, Freeze" — catabolic emergency response	"Rest and Digest" — anabolic restorative functions

Effector Organ	Sympathetic Effect (NE)	Parasympathetic Effect (ACh)
Heart	↑ HR (β_1 — SA node); ↑ Contractility (β_1); ↑ Conduction velocity (β_1)	↓ HR (M2 — SA node: ↑ K^+ conductance); ↓ AV conduction (M2)
Coronary arteries	Dilate (β_2) in physiological stress; constrict (α_1)	Dilate (minor direct effect via M3 on endothelium → NO)
Bronchial smooth muscle	Relax/Dilate (β_2 — bronchodilation)	Constrict (M3 — bronchoconstriction); ↑ secretion
GI smooth muscle	Inhibit motility (α_2 , β_2); close sphincters (α_1)	Stimulate peristalsis (M3); relax sphincters; ↑ secretion
Urinary bladder	Relax detrusor (β_2); contract internal sphincter (α_1) — urine retention	Contract detrusor (M3); relax internal sphincter — urination
Pupils	Dilate (α_1 → mydriasis)	Constrict (M3 → miosis)
Salivary glands	Scanty, viscous secretion (α_1)	Copious, watery secretion (M3)
Sweat glands	Increase sweating (cholinergic sympathetic — M receptors)	No significant innervation
Blood vessels (skin/splanchnic)	Constrict (α_1)	Little direct innervation; relax via local NO
Blood vessels (skeletal muscle)	Dilate (β_2); constrict (α_1)	Dilate (NO-mediated)
Liver	Glycogenolysis; gluconeogenesis (β_2 , α_1)	Minor effects
Adipose tissue	Lipolysis (β_1/β_3 — HSL activation)	Minor effects

§12 REFLEXES

Reflex	Arc	Afferent	Efferent	Clinical Note
Stretch reflex (Deep Tendon)	Monosynaptic; spinal segment	Ia afferent (muscle spindle)	α -motor neuron (same muscle)	Knee jerk (L3/4); ankle jerk (S1); biceps (C5/6); triceps (C7). Absent = LMN lesion; Hyperreflexic = UMN lesion
Golgi Tendon Organ (inverse stretch)	Disynaptic (Ib inhibitory interneuron)	Ib afferent (tendon)	Inhibits α -MN of same muscle; facilitates antagonist	Clasp-knife response in spasticity

Reflex	Arc	Afferent	Efferent	Clinical Note
Withdrawal (Flexor) reflex	Polysynaptic; multi-segmental	A-6, C pain afferents	Flexors contract; extensors inhibited (ipsilateral); crossed extension reflex (contralateral)	Protective reflex from noxious stimuli; exaggerated in UMN lesions
Babinski reflex	Polysynaptic	Plantar skin	Hallux dorsiflex; fanning of toes	Normal in infants; ABNORMAL (UMN sign) after age 2
Pupillary light reflex	Polysynaptic; midbrain	CN II (optic — afferent)	Edinger-Westphal → CN III → constrictor pupillae (M3)	Tests CN II and III; direct and consensual; RAPD = afferent defect
Corneal reflex	Polysynaptic; pons/medulla	CN V1 (nasociliary branch)	CN VII (facial — orbicularis oculi)	Tests CN V and VII; absent in CN V1 or VII lesions; last brainstem reflex to go with herniation
Accommodation reflex	Cortical + midbrain	CN II + cortex (occipital)	CN III: convergence, pupil constriction, ciliary muscle → lens ↑ curvature	Tests CN III; light-near dissociation in Argyll Robertson pupil (neurosyphilis — no light, yes near)